RECOVERY RUNBOOK

VIRTUAL CONTROL

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Appliance Recovery — Read-Only FS & kubelet/DNS

Recovering a VCSA from EXT4 aborted-journal read-only, and an Aria Automation appliance from a kubelet failure caused by a corrupted /etc/hosts.

VCSA | EXT4 | FSCK | ARIA AUTOMATION | KUBELET | /ETC/HOSTS

v1.0 VMware Cloud Foundation 9

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VCF Appliance Recovery Runbook — Read-Only Filesystem & kubelet/DNS Recovery

Prepared by: Virtual Control LLC **Date:** June 22, 2026 **Document Version:** 1.0 **Classification:** Internal

Scope. Two reusable appliance-recovery procedures, captured from a live incident on **June 22, 2026** in which a home-network change (re-cabling and a router/subnet move during the lab-isolation cutover) caused brief guest I/O / DNS disruption. Two management appliances came up unhealthy afterward — **vCenter** (web UI dead) and **Aria Automation** (platform dead). The underlying vSAN/storage and all other appliances were healthy throughout; both failures were inside the appliance guests. Procedure A recovers a VCSA whose filesystems went read-only from an EXT4 aborted journal. Procedure B recovers an Aria Automation node whose kubelet won't start because `/etc/hosts` is corrupted/incomplete.

Companion documents. *Lab Network Isolation Runbook* (the cutover that preceded this); *VCF9 Operations Campaign 2026-06-22*.

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1. Triage — Which Procedure Do I Need?

Observation	Likely state	Procedure
Appliance answers **ping**, web UI / service **port down**; SSH works but commands may hang	Guest is up but a service or the filesystem is wedged	continue triage

Observation	Likely state	Procedure
Console shows `EXT4-fs error … Detected aborted journal` / `Remounting filesystem read-only`	Filesystem(s) read-only from an I/O event	**A — VCSA EXT4 recovery**
`mount` shows root as `ro`, writes fail with `Read-only file system`	Root filesystem read-only	**A** (or remount + B if writable)
Appliance is a k8s-based product (Aria Automation/vRA), console shows `[FAILED] Failed to start kubelet`, disk **not** full	kubelet can't reach the API server — usually name resolution	**B — kubelet /etc/hosts recovery**

First, rule out storage. Before rebooting any appliance, confirm the underlying datastore is healthy so you are not rebooting onto failing storage. For vSAN: `esxcli vsan debug object health summary get` (expect 0 inaccessible) and `esxcli vsan cluster get` (member count correct). For local VMFS: confirm free space and no host disk errors. In this incident vSAN was 4/4, 213 objects healthy — so both failures were guest-internal, and a reboot/repair was safe.

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2. Procedure A — VCSA EXT4 Read-Only Recovery

Symptom. vCenter UI (443) unreachable while ping/SSH respond; the VM console shows EXT4 aborted-journal errors and filesystems remounted read-only. As the read-only condition spreads, SSH may begin refusing connections.

Root cause. A transient guest I/O stall (storage latency, a brief disconnect, or a VM reset) causes EXT4 to abort its journal and **remount the affected volumes read-only** as a protection. vCenter services (`vpxd`, `vsphere-ui`) cannot run on read-only storage, so the UI dies. This is **not** a storage failure when the datastore underneath is healthy.

Fix — reset the VM so Photon runs a filesystem check on boot.

1. Confirm storage is healthy (Section 1 info-box).
2. The appliance cannot repair in place (filesystems are read-only). **Reboot it.** Tools is usually unresponsive, so a graceful guest reboot fails — issue a **reset** from the ESXi host: `bash # on the ESXi host running the VCSA; get the vmid first vim-cmd vmsvc/getallvms | grep -i vcenter vim-cmd vmsvc/power.reset <vmid>` The read-only filesystems have no in-flight writes, so a reset does not worsen on-disk state.
3. On boot, Photon **replays the EXT4 journal / runs fsck** and remounts the volumes read-write. If the boot stalls at `(initramfs)` / emergency mode / "*failed to mount /sysroot*", add `fsck.repair=yes` to the GRUB kernel line (KB 326323) and boot.

4. Allow **~10–15 minutes** for all services to start (the UI is last).

Verify the recovery (don't trust the port alone)

A TCP-443 check can pass early (envoy binds 443 before the UI is ready). Confirm the real state via SSH (drop to `shell` from `appliancesh`):

```
# all filesystems read-write? (empty result = good)
mount | grep ext4 | grep -E '[ (]ro[ ,)]'
# recent EXT4 state — want clean "mounted filesystem", no errors
dmesg | grep -iE 'EXT4-fs|journal|read-only' | tail
# services
service-control --status
```

Expect `vmware-vpxd`, `vsphere-ui`, `vmware-vsan-health` **Running**. The following are **normally Stopped** (optional features) and are **not** a problem: `vmcam`, `vmware-imagebuilder`, `vmware-rbd-watchdog`, `vmware-netdumper`, `vmware-vcha`.

Watch-item. If the appliance goes read-only **again** after recovery, the transient stall was not a one-off — investigate the datastore the VM lives on (vSAN latency, or a local disk's SMART/health).

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3. Procedure B — Aria Automation kubelet / `/etc/hosts` Recovery

Symptom. Aria Automation appliance answers ping and its SSH port is open, but the platform (443) is down and SSH *commands* hang (login shell blocked). The VM console shows **`[FAILED] Failed to start kubelet: The Kubernetes Node Agent`**. Disk is **not** full.

Root cause chain (as diagnosed live).

1. The same network disruption left the appliance's **root filesystem mounted read-only** — blocking SSH login (can't write session files) and any config change.
2. `/etc/hosts` was **corrupted into binary** during the read-only/I-O event. `getent` (glibc) tolerates the garbage and still resolves, but **kubelet is written in Go and cannot parse a corrupted `/etc/hosts`** — so it ignores the file and falls back to DNS, which fails. kubelet's log shows `dial tcp: lookup <name>: no such host`.
3. kubelet reaches the Kubernetes API at **`vra-k8s.local:6443`** (not the appliance FQDN). A clean rewrite of `/etc/hosts` must include **both** the appliance FQDN **and** `vra-k8s.local` mapped to the node IP, or kubelet still can't find the API server.

Fix. Work from the **VM console** (SSH login is blocked while the fs is read-only).

1. Remount root read-write:

```
mount -o remount,rw /
# verify writable
touch /etc/.wtest && echo WRITABLE && rm /etc/.wtest
```

(If `remount,rw` errors, the fs needs an offline fsck — boot to a root shell via GRUB `init=/bin/bash`, `fsck -y /dev/sdaN`, reboot.)

2. Confirm `/etc/hosts` is corrupt and find the API name kubelet needs:

```
file /etc/hosts                      # "data"/binary instead of "ASCII text" = corrupt
grep 'server:' /etc/kubernetes/*.conf   # shows  server: https://vra-k8s.local:6443
```

3. Rewrite `/etc/hosts` as clean text — include the FQDN *and* `vra-k8s.local`:

```
cp /etc/hosts /etc/hosts.corrupt.bak
printf '127.0.0.1 localhost\n::1 localhost ip6-localhost ip6-loopback\n<NODE_IP> <fqdn>
<short>\n<NODE_IP> vra-k8s.local\n' > /etc/hosts
file /etc/hosts                      # must now say "ASCII text"
getent hosts vra-k8s.local           # must resolve to <NODE_IP>
```

(Use the appliance's own management IP for `<NODE_IP>`; the API server listens on `*:6443`, and the cert validates on the name, so the node IP is the correct, standard mapping.)

4. Restart kubelet (non-blocking — the unit's start-wait can exceed the systemd timeout):

```
systemctl restart kubelet --no-block
sleep 30
systemctl is-active kubelet          # want: active (or activating, then active)
journalctl -u kubelet -n 15 --no-pager
```

What success looks like

The `no such host` and `node "<fqdn>" not found` errors **stop**, replaced by normal kubelet chatter (`pod_container_deletor`, `Throttling request`, container create/delete). kubelet registers the node, the control-plane static pods (API server already listening on `:6443`) are reached, and the Aria pods come up over **~10–15 minutes** until 443 serves.

Why `getent` "worked" but kubelet didn't. glibc's resolver (used by `getent`, `ping`, curl) skips unparseable lines in `/etc/hosts`; Go's resolver (kubelet) treats a non-text `/etc/hosts` as unusable and falls back to DNS. A corrupted `/etc/hosts` therefore breaks Go services while leaving shell tools looking fine — always check `file /etc/hosts` when a Go-based service reports `no such host` for a name that `getent` resolves.

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4. Document Information

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Distribution	Lab notebook

Revision History

Version	Date	Notes
1.0	2026-06-22	Initial release. Captured live from the June 22 incident: VCSA EXT4 read-only recovery (Procedure A) and Aria Automation kubelet `/etc/hosts` recovery (Procedure B), including the Go-vs-glibc `/etc/hosts` parsing insight and the `vra-k8s.local` API-endpoint requirement.

Document End

See also:

- Lab Network Isolation Runbook (the network change that triggered this)
- KB 326323 — Photon initramfs fsck recovery (`fsck.repair=yes`)
- VCF9 Operations Campaign 2026-06-22-to-2026-06-26